



Meguro MSG-2051A GPS signal generator

know his correct location.

The objective of jamming is to simply interrupt the availability of GPS signals in space at the receiver. In-band or out-of-band harmonic RF transmissions could mask the weak GPS spread-spectrum signals. The effect is to corrupt the signal at the receiver so that no valid GPS signal can be decoded by the receiver.

Spoofing is more serious than jamming because jamming causes the service to degrade in performance, while spoofing takes control of the user receiver. Jamming is a low-tech method and at best, a prank. It would not serve the purpose of terrorists or those assisting them. They must mislead the agencies intercepting the calls, which falls in the realm of spoofing and could be done in two ways: misleading the GPS receiver and misleading the location server.

Misleading the GPS receiver

This involves an attacker providing the receiver with a misleading signal, fooling the receiver to use fake signals in space for positioning calculations. The receiver produces a misleading position solution.

Two basic configurations are possible: GPS signal generator and GPS receiver spoofers.

GPS signal generator. GPS signal generators that are readily available

from several vendors could be used as spoofers. For use as a spoofer, the RF output of the signal generator is amplified and transmitted, possibly using a directional antenna. In this case, the transmitted signals are not phase- and frequency-matched to the GPS signals being received from satellites in the locality, and the navigation data does not replicate the currently active navigation data.

Although a receiver could be fooled by this approach, particularly if the target receiver is first jammed and forced to reacquire, the spoofing signal generated in this fashion typically looks like noise—rather than a usable signal—to a receiver tracking it.

GPS receiver spoofer. Spoofers in this category are coupled to a GPS receiver. The GPS receiver tracks satellite signals at a location and decodes the navigation data. The spoofer then generates a signal that mimics the incident satellite signals in all respects. Conceivably, a spoofer could add a calculated offset to each satellite signal to compensate for a specified geometric offset to the target GPS antenna. The spoofer is also able to vary the strength of the constituent signals so that they appear at the target antenna having the same relative strengths as the authentic signals.

Cornell University's GRID dual-frequency software-defined GPS receiver is an example of this type of spoofer.

It can simultaneously track twelve coarse/acquisition (C/A) channels and generate eight C/A spoofing channels. The hardware required for this type of spoofer is neither very costly nor difficult to procure but the technical knowledge required must be of a high order. The Cornell experiments were widely reported in the media and led to a great deal of speculations.

There are four GPS observables that can be directly measured by a GPS receiver: the GPS message, code ranges, fractional phase ranges and Doppler shift. In a GPS receiver, pseudorandom noise (PRN) code correlation is performed at a high frequency to remove the PRN code. The received GPS signal is passed through a high-pass filter to remove navigation data. After the navigation message is removed, the resulting signal is the Doppler-shifted carrier. The Doppler-shifted carrier is then passed to a phase-locked loop (PLL) and compared with a receiver-generated carrier to get the fractional phase offset.

The GPS message can be spoofed because PRN code is available to the public. The PRN code ranges of C/A code and P-code are direct observables. One chip of C/A code range will cause 300km ambiguity. The range measurement for C/A code is the basic observation of GPS receivers and it is derived from the time shift of the receiver PRN sequence and the PRN sequence that is multiplexing the incoming signal. This time shift, however, can be controlled by spoofing the transmitter. Pseudo-range measurements can then be modeled. The satellite position vector can be forged by spoofing the GPS messages. Using this model, the PRN code shift can be easily calculated.

Spoofing a static receiver (called static spoofing) is much easier than spoofing a dynamic receiver. For a stationary receiver, the position solution is fixed and the spoofed pseudo-range can be calculated using code range model. For a moving receiver (called dynamic spoofing), if the spoofer does